Lecture 2: Costs and Processing

Vertical Slices, MVPs, Crawling, Walking, and Running

University of Chicago

April 2, 2026

Outline

1. Readings
2. Development Methodology
3. AI Resume Scoring System

Outline

1. Readings
2. Development Methodology
3. AI Resume Scoring System

Readings Overview

1. **Context Engineering** - Understanding how to design effective prompts and context
2. **Gary Marcus on AI Limitations** - Critical analysis of what AI can and cannot do
3. **Gas Town** - Broader perspective on AI development (optional, but interesting!)

Context Engineering

Reading 1: Effective Context Engineering for AI Agents

- **Source**: Anthropic
- Some design information and breakdown of what goes into context

Reading 2: Context Engineering

- **Source**: Simon Willison
- Great blog on what is happening in AI and developmen

Reading 3: Agent Best Practices

- **Source**: Cursor
- Similar to Anthropic Article

Gas Town

Welcome to Gas Town by Steve Yegge

- **Length**: LONG essay
- Below are the stages of using AI for coding and development. Which are you?!?

Stage 1	Zero or Near-Zero AI: maybe code completions, sometimes ask Chat questions
Stage 2	Coding agent in IDE, permissions turned on. A narrow coding agent in a sidebar asks your permission to run tools.
Stage 3	Agent in IDE, YOLO mode: Trust goes up. You turn off permissions, agent gets wider.
Stage 4	In IDE, wide agent: Your agent gradually grows to fill the screen. Code is just for diffs.

Gas Town (continued)

Stage 5	CLI, single agent. YOLO. Diffs scroll by. You may or may not look at them.
Stage 6	CLI, multi-agent, YOLO. You regularly use 3 to 5 parallel instances. You are very fast.
Stage 7	10+ agents, hand-managed. You are starting to push the limits of hand-management.
Stage 8	Building your own orchestrator. You are on the frontier, automating your workflow.

Gary Marcus on AI

Reading 4: *Why ChatGPT Can't Be Trusted With*

Marcus on AI

Why ChatGPT can't be trusted with breaking news

A new case in point

GARY MARCUS
JAN 03, 2026

329 134 43 Share

Reading 5: *Let's Be Honest: Generative AI Isn't*

Marcus on AI

Let's be honest, Generative AI isn't going all that well

A sampling of recent news

GARY MARCUS
JAN 12, 2026

352 184 38 Share

Some recent news, all long anticipated by this newsletter:

1. LLMs can still cannot be trusted:

Outline

1. Readings
2. Development Methodology
3. AI Resume Scoring System

System Thinking: Breaking Down AI Tasks

- Before coding, we need to **think systematically**:
  1. How do we abstract and organize our code?
  2. How do we break tasks into inputs and outputs?
  3. How do we organize these into functions?
- **Plan before you code**
- Break down required tasks, *then* start coding
- How do we do this? What does it mean to "break tasks down"?

The Development Process

Three-Step Approach:

1. **Define a vertical slice**
   - What is the minimal end-to-end flow?
   - Start simple, prove it works
2. **Define the systems (functions) required**
   - Know the inputs and outputs of each piece
   - Simpler is better
3. **Then: Crawl $\rightarrow$ Walk $\rightarrow$ Run**
   - Start minimal, add features incrementally

Example Problem: Time Series Analysis

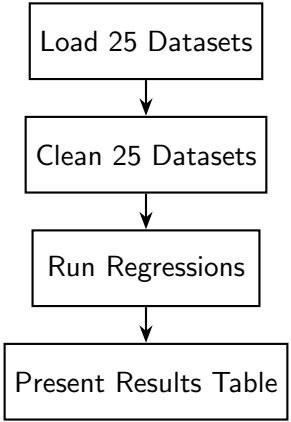
Task: Run a regression on 25 datasets

Context:

- We have data from 25 countries
- Each CSV contains education and GDP data over time
- We want to run a time series analysis on the relationship

Question: How do we break this down?

System Design: Complete Problem



This is the **complete problem**, but where do we start?

Inputs and Outputs: The Missing Piece

For each system component, we need to define:

- **Load Data**: What format for each column? How to handle dates? Any transformations?
- **Clean Data**: What cleaning steps? How do we handle missing values?
- **Run Regression**: What type? Linear? Time series?
- **Present Output**: Table? Graph? JSON? What metrics?

Inputs and Outputs: The Missing Piece

For each system component, we need to define:

- **Load Data**: What format for each column? How to handle dates? Any transformations?
- **Clean Data**: What cleaning steps? How do we handle missing values?
- **Run Regression**: What type? Linear? Time series?
- **Present Output**: Table? Graph? JSON? What metrics?

This is the complete problem, but we never want to start here!

We want to start with a **vertical slice**.

What is a Vertical Slice?

A vertical slice:

- Contains **everything** needed to go from top to bottom
- Works on a **subset** of inputs
- Proves the **end-to-end flow** works
- Is the **simplest possible** complete solution

When we build AI systems, we want to start with vertical slices.

Why Vertical Slices for AI?

AI systems have unique challenges:

- **Complex**: Many moving parts and dependencies
- **Uncertain failure points**: We aren't sure where things will break
- **Non-deterministic**: Errors appear in unexpected places
- **Iterative prompting**: Prompts need testing and refinement

With vertical slices:

- We build a viable end-to-end solution that works on a **subset** of inputs
- We discover integration issues **early**
- We can demonstrate **value quickly**

The Opposite: Horizontal Development

Horizontal development means:

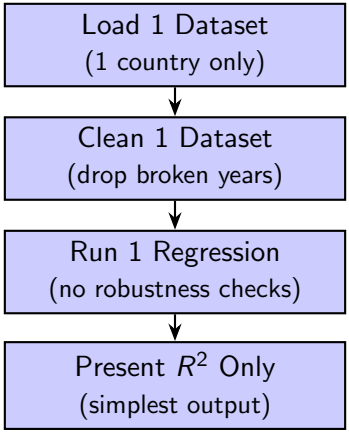
- Building for **all contingencies** at each level
- Handling **every possibility** before moving forward
- Creating **complete infrastructure** before testing end-to-end

For our regression example:

- Load all 25 datasets with full error handling
- Clean all datasets with every edge case covered
- Build multiple regression types and robustness checks
- Create multiple output formats and visualizations

Problem: You spend weeks building infrastructure before knowing if the core idea works!

Vertical Slice: Regression Example



This proves it works end-to-end with minimal effort!

Example: Personalized Marketing Emails

Task: Send AI-generated personalized marketing emails to leads

Vertical Slice Approach:

- Load **one** lead from the database
- Generate **one** email with LLM
- Print email to console for manual review
- If it looks good, move to next step

Horizontal Approach (what NOT to do):

- Build complete database integration for all leads
- Create email templates for all scenarios
- Build full SMTP sending infrastructure
- Set up monitoring and analytics before sending anything

Principle: Well-Defined Inputs and Outputs

- Each component should have:
  1. **Clear input contract**: What data does it expect? In what format?
  2. **Clear output contract**: What data does it return? In what format?
  3. **Single responsibility**: Does one thing well
- **Benefits:**
  - Easy to understand what each piece does
  - Easy to test each piece independently
  - Easy to replace or improve individual pieces
  - Clear error boundaries

Methodology: Crawl, Walk, Run

- **Crawl**: What is the minimum code to verify it's possible?
  - Hardcoded examples
  - Manual verification
  - Proof of concept that it can work in *some* instances
- **Walk**: What is the minimum code to verify it works in most cases?
  - Handle common edge cases
  - Basic error handling
  - Works reliably for typical inputs
- **Run**: What does the full production system look like?
  - Comprehensive error handling
  - Monitoring and logging
  - Scale and performance optimization

Crawl Phase in Detail

Goal: Prove the concept works at all

Characteristics:

- Use **hardcoded** or sample data
- **No error handling** (let it fail if something breaks)
- **Manual testing** (print statements, visual inspection)
- **Happy path only** (assume everything works)

Example - Resume Screening:

- Hardcode path to one resume PDF
- Extract skills with LLM, print to console
- Compare to hardcoded job requirements
- Print match score

If this doesn't work, nothing else matters!

Building Vertical Slices First

- **Vertical Slice**: End-to-end functionality for a narrow use case
  - Example: Process ONE resume through the entire pipeline
  - Proves the concept works end-to-end
  - Surfaces integration issues early
- **Then Build Horizontally**: Expand capabilities
  - Add batch processing
  - Add error handling
  - Add more sophisticated matching
  - Add caching and optimization

Key Takeaways: Building with AI

1. **Define the vertical slice** – what is the minimum thing you can get working?
   - Prove end-to-end flow works with simplest case
   - Identify integration issues early
2. **Figure out the inputs and outputs**
   - Each function needs clear contracts
   - Simpler is better
3. **Figure out the pricing**
   - Know your costs before scaling
4. **Expand horizontally after vertical slice works**
   - Add batch processing, error handling, features
   - Don't optimize or add features prematurely

Outline

1. Readings
2. Development Methodology
3. AI Resume Scoring System

Problem Background

- We have been hired by a company to help with their software developer hiring process
- Our goal is to build an AI-assisted resume scoring system
- For each resume, we want to score it so that we don't waste time on resumes we are unlikely to hire
- We are also cost-conscious.
- Scores should be on a 0-100 scale.

Brainstorming

- How are we going to proceed here?
- What's our vertical slice?
- What are the system components?
- What are the inputs and outputs?
- How much will our decisions cost?

Let's Look at Our Inputs

- **Data**: `resumes_final.csv` has one row per resume (anonymized, taken from Kaggle)
- **Functions**: Our notebook has two functions already:
  1. `load_resumes()` - loads all resumes into a dictionary
  2. `analyze_resume()` - runs a prompt against a resume with structured output
- There is a job req in the directory: `job_req_senior.md`

Define our system

- Do we have a good definition of each component?
- Do we know the inputs and outputs and what has to happen inside each system?

Next Step: Crawling!
Start with the simplest possible version

Resume Screening System: Crawl, Walk, Run

Phase 1: Vertical Slice (Crawl)

- Generate score for a single resume for a single job req

Phase 2: Horizontal Expansion (Walk)

- Generate score for **most** resumes for a single job req

Phase 3: Production (Run)

- Work on **all** resumes for **all** job reqs
- Optimize prompts and costs
- Add caching for repeated skills
- Add human-in-the-loop review
- Monitor accuracy and performance

Next Steps: Hands-On Practice

Goal: Build a working resume scorer

Your Task:

1. **Form teams** (2-3 people)
2. **Open the notebook**: `resume_screening.ipynb`
3. **Work through the examples**:
   - Load resume data
   - Use structured outputs to extract information
   - Match resumes to job requirements
4. **Key questions to answer**:
   - Can we create a score (0-100)?
   - How do we do that?
   - What prompt and schema do we need?
   - How much does it cost per resume?

Focus on the Crawl First!

Start simple, prove it works